

2024, Mar. 2024 Release LOC:Acute Adult

Arrhythmia, Atrial

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ⁽¹⁾Episode Day 2 ⁽³⁸⁾Episode Day 3 ⁽⁵⁷⁾

Episode Day 4

Discharge Screens ⁽⁵⁹⁾

Utilization Benchmarks

Notes

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Overview

Instruction: This subset is for patients who present with atrial arrhythmias and require evaluation and treatment. For ventricular arrhythmias see the Arrhythmia: Ventricular or abnormal ECG subset. For heart blocks see the Arrhythmia: Blocks subset. Criteria for the following atrial arrhythmias are found within this subset:

- Atrial fibrillation (AF)
- Atrial flutter (AFL)
- Supraventricular tachycardia (SVT)
- Bradycardia

Introduction:

Arrhythmias are often triggered by alcohol, myocardial infarction (MI), pericarditis, myocarditis, pulmonary embolism, pulmonary disease, hyperthyroidism, metabolic disease, electrolyte imbalance, or cardiac drug toxicity. Presenting symptoms of an arrhythmia often include: palpitations, chest pain, dyspnea, fatigue, lightheadedness, or syncope.

Evaluation/Intervention:

Patients presenting with an arrhythmia should undergo a thorough history and physical examination. A 12-lead ECG is recommended in all patients being evaluated for an arrhythmia. Additional testing may be necessary and includes: left ventricular functional assessment using transthoracic echocardiogram, computed tomography (CT), cardiac magnetic resonance imaging (MRI), electrophysiology study (EPS), and laboratory testing.

Treatment goals for patients with an arrhythmia may include modifying and preventing risk factors, restoration to sinus rhythm, rate control, direct-current cardioversion, prevention of thromboembolism, and the prevention of future arrhythmias. Recent guidelines for atrial fibrillation highlights importance of identifying and treating risk factors and instituting lifestyle changes to decrease the likelihood of developing AF. If comorbidities are present, those would also require an evaluation (Joglar et al., J Am Coll Cardiol 2023;). For all atrial arrhythmias, management may include discontinuation of proarrhythmic medication, initiation of antiarrhythmic medication, immunotherapy with digoxin specific antibody, insertion of an implantable cardioverter defibrillator (ICD) or pacemaker, catheter ablation, or surgery. For atrial fibrillation in particular, care for patients must include stroke assessment and appropriate therapies, symptom management with rate or rhythm control and risk factor modifications (Joglar et al., J Am Coll Cardiol 2023;; January et al., Circulation 2019, 140: e125-e51; Brignole et al., Eur Heart J 2018, 39: 1883-948; Chen et al., Circulation 2018, 137: e623-e44; January et al., Circulation 2014, 130: 2071-104).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most

appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽³⁾● New onset atrial fibrillation or flutter and, **All:** ^(4, 5, 6, 7)

– Heart rate < 110/min post initial antiarrhythmic treatment (includes PO)

– Systolic blood pressure ≥ 100 mmHg post initial antiarrhythmic treatment ⁽⁸⁾● Intervention, ≥ **One:**

– Electrophysiology (EP) or cardiology consult ≤ 24h

– Initiation or dose adjustment of antiarrhythmic (includes PO)

– Electrical cardioversion performed ≤ 24h ⁽⁹⁾● Atrial fibrillation or flutter and heart rate < 110/min post initial antiarrhythmic (includes PO) and, ≥ **One:** ^(5, 10, 11)

– Electrophysiology (EP) or cardiology consult ≤ 24h

– Initiation or dose adjustment of antiarrhythmic (includes PO)

– Electrical cardioversion performed ≤ 24h ⁽⁹⁾– Resolved atrial fibrillation or flutter post ibutilide administration ⁽¹²⁾● **INTERMEDIATE, ≥ One:** ⁽¹³⁾● Atrial fibrillation or flutter and heart rate ≥ 110/min post initial antiarrhythmic (includes PO) and, **One:** ^(4, 5, 6, 10)

– Antiarrhythmic

– Digoxin loading ≤ 24h ⁽¹⁴⁾– Permanent pacemaker placement within last 24h ⁽¹⁵⁾● Atrial fibrillation and acute heart failure, **All:** ⁽¹⁶⁾– NYHA Class III or IV ⁽¹⁷⁾

– Antiarrhythmic

– Diuretic

● Supraventricular tachycardia and, ≥ **One:** ⁽¹⁸⁾– Antiarrhythmic ⁽¹⁹⁾– Electrical cardioversion ⁽²⁰⁾– Bradycardia and symptomatic ^(21, 22, 23)● Suspected drug toxicity and bradycardia and, **Both:** ⁽²³⁾– Symptomatic ^(21, 22)

– Continuous cardiac monitoring (excludes Holter)

● Initiation or dose adjustment of antiarrhythmic (includes PO) with proarrhythmic risk ≤ 3d and, ≥ **One:** ^(24, 25, 26)– Dofetilide ⁽²⁷⁾

– Sotalol

● **CRITICAL, ≥ One:** ⁽²⁸⁾● Bradycardia and, ≥ **One:** ^(21, 23)– Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾

– Dopamine

– Epinephrine

– Isoproterenol ⁽³⁰⁾– Temporary pacemaker ≤ 2d ⁽³¹⁾● Hemodynamic instability (excludes bradycardia), **Both:**● Finding, ≥ **One:**● Systolic blood pressure, ≥ **One:**

– < 90 mmHg

– < 110 mmHg with chronic hypertension

– > 30 mmHg decrease from baseline ⁽³²⁾– Labile ⁽³³⁾

- MAP < 65 mmHg
- **Intervention, ≥ One:**
 - Antiarrhythmic
 - Electrical cardioversion ^(9, 20)
 - Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾
- Hemodynamic monitoring, invasive ⁽³⁴⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** ⁽¹⁾
(Symptom or finding within 24h)
- **OBSERVATION, One:** ⁽³⁾
 - **New onset atrial fibrillation or flutter and, All:** ^(4, 5, 6, 7)
 - Heart rate < 110/min post initial antiarrhythmic treatment (includes PO) ⁽⁸⁾
 - Systolic blood pressure ≥ 100 mmHg post initial antiarrhythmic treatment (includes PO) ⁽⁸⁾
 - **Intervention, ≥ One:**
 - Electrophysiology (EP) or cardiology consult ≤ 24h
 - Initiation or dose adjustment of antiarrhythmic (includes PO)
 - Electrical cardioversion performed ≤ 24h ⁽⁹⁾
 - **Atrial fibrillation or flutter and heart rate < 110/min post initial antiarrhythmic (includes PO) and, ≥ One:** ^(5, 10, 11)
 - Electrophysiology (EP) or cardiology consult ≤ 24h
 - Initiation or dose adjustment of antiarrhythmic (includes PO)
 - Electrical cardioversion performed ≤ 24h ⁽⁹⁾
 - Resolved atrial fibrillation or flutter post ibutilide administration ⁽¹²⁾
- **ACUTE, One:** ⁽³⁵⁾
 - Heart failure (**use Heart Failure subset**)
 - **Comorbid atrial fibrillation or flutter in a hospitalized patient and, One:** ^(36, 37)
 - **New onset atrial fibrillation or flutter and, All:** ^(4, 5, 6)
 - Heart rate < 110/min post initial antiarrhythmic treatment (includes PO) ⁽⁸⁾
 - Systolic blood pressure ≥ 100 mmHg post initial antiarrhythmic treatment (includes PO)
 - **Intervention, ≥ One:**
 - Electrophysiology (EP) or cardiology consult ≤ 24h
 - Initiation or dose adjustment of antiarrhythmic (includes PO)
 - Electrical cardioversion performed ≤ 24h ⁽⁹⁾
 - **Atrial fibrillation or flutter and heart rate < 110/min post initial antiarrhythmic (includes PO) and, ≥ One:** ^(5, 10, 11)
 - Electrophysiology (EP) or cardiology consult ≤ 24h
 - Initiation or dose adjustment of antiarrhythmic (includes PO)
 - Electrical cardioversion performed ≤ 24h ⁽⁹⁾
 - Resolved atrial fibrillation or flutter post ibutilide administration ⁽¹²⁾
- **INTERMEDIATE, ≥ One:** ⁽¹³⁾
 - **Diagnosis, actual or suspected, One:**
 - Acute coronary syndrome (**use Acute Coronary Syndrome subset**)
 - Fractured or displaced pacemaker lead (**use General Medical subset**)
 - Implantable device failure (**use General Medical subset**)
 - **Atrial fibrillation or flutter and heart rate ≥ 110/min post initial antiarrhythmic (includes PO) and, One:** ^(4, 5, 6, 10)
 - Antiarrhythmic
 - Digoxin loading ≤ 24h ⁽¹⁴⁾
 - Permanent pacemaker placement within last 24h ⁽¹⁵⁾
 - **Atrial fibrillation and acute heart failure, All:** ^(4, 16)
 - NYHA Class III or IV ⁽¹⁷⁾
 - Antiarrhythmic
 - Diuretic
 - **Supraventricular tachycardia and, ≥ One:** ⁽¹⁸⁾

- Antiarrhythmic ⁽¹⁹⁾
- Electrical cardioversion ⁽²⁰⁾
- Bradycardia and symptomatic and, ≥ **One:** ^(21, 22, 23)
 - Atropine
 - Permanent pacemaker placement within last 24h ⁽¹⁵⁾
- Suspected drug toxicity and bradycardia and, **Both:** ⁽²³⁾
 - Symptomatic ^(21, 22)
 - Continuous cardiac monitoring (excludes Holter)
- Initiation or dose adjustment of antiarrhythmic (includes PO) with proarrhythmic risk ≤ 3d and, ≥ **One:** ^(24, 25)
 - Dofetilide ⁽²⁷⁾
 - Sotalol
- **CRITICAL, ≥ One:** ⁽²⁸⁾
 - Bradycardia and, ≥ **One:** ^(21, 23)
 - Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾
 - Dopamine
 - Epinephrine
 - Isoproterenol ⁽³⁰⁾
 - Temporary pacemaker ≤ 2d ⁽³¹⁾
 - Hemodynamic instability (excludes bradycardia), **Both:**
 - Finding, ≥ **One:**
 - Systolic blood pressure, ≥ **One:**
 - < 90 mmHg
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽³²⁾
 - Labile ⁽³³⁾
 - MAP < 65 mmHg
 - Intervention, ≥ **One:**
 - Antiarrhythmic
 - Electrical cardioversion ^(9, 20)
 - Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾
 - Hemodynamic monitoring, invasive ⁽³⁴⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ⁽³⁸⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Arrhythmia controlled or resolved ^(40, 41)
 - Medication regimen established and tolerated
 - Vital signs within acceptable limits
 - **Partial Responder**, not clinically stable for discharge and requires continued stay, **One:**
 - Antiarrhythmic medication adjustment (includes PO) ⁽⁴²⁾
 - **ACUTE, One:** ⁽³⁵⁾
 - **Early Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Arrhythmia controlled or resolved ⁽⁴¹⁾
 - Medication regimen established and tolerated
 - Vital signs within acceptable limits
 - No complication or active comorbidity relevant to this episode of care ⁽⁴³⁾
 - **Partial Responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Antiarrhythmic medication adjustment (includes PO) ⁽⁴²⁾
 - Unfractionated heparin, continuous and, ≥ **One:** ⁽⁴⁴⁾
 - Antiphospholipid antibody syndrome ⁽⁴⁵⁾
 - Bleeding, active or increased risk
 - Creatinine clearance ≤ 30 mL/min ⁽⁴⁶⁾

- BMI ≥ 40 kg/m² or weight > 120kg ⁽⁴⁷⁾
- Heart failure and, **Both:** ⁽⁴⁸⁾
 - Symptom or finding, $\geq$ **One:** ⁽³²⁾
 - O₂ sat $\leq 91\%$ (0.91) and < baseline ⁽³²⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽⁴⁹⁾
 - Intervention, $\geq$ **One:** ⁽⁵⁰⁾
 - Diuretic, **One:** ⁽⁵¹⁾
 - $\geq 2x/24h$
 - Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline ^(32, 42)
 - Dialysis
- Post Critical care $\leq 24h$
- **INTERMEDIATE, $\geq$ One:**
 - Atrial arrhythmia and, $\geq$ **One:** ⁽⁵²⁾
 - Antiarrhythmic
 - Conversion from IV to PO antiarrhythmic with no proarrhythmic risk $\leq 24h$
 - Conversion from IV to PO antiarrhythmic with proarrhythmic risk $\leq 3d$ ^(26, 53)
 - Bradycardia and, **Both:** ⁽²³⁾
 - Atropine
 - Symptomatic ^(21, 22)
 - Suspected drug toxicity and bradycardia and, **Both:** ⁽²³⁾
 - Symptomatic ^(21, 22)
 - Continuous cardiac monitoring (excludes Holter)
 - NIPPV ⁽⁵⁴⁾
 - Oxygen $\geq 40\%$ (0.40), $\leq 2d$ ⁽⁵⁵⁾
 - Permanent pacemaker placement within last 24h ⁽¹⁵⁾
 - Post cardioversion and post Critical care $\leq 24h$
 - Initiation or dose adjustment of antiarrhythmic (includes PO) with proarrhythmic risk $\leq 3d$ and, $\geq$ **One:** ^(24, 25, 26)
 - Dofetilide ⁽²⁷⁾
 - Sotalol
- **CRITICAL, $\geq$ One:**
 - Bradycardia and, $\geq$ **One:** ^(21, 23)
 - Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾
 - Dopamine
 - Epinephrine
 - Isoproterenol ⁽³⁰⁾
 - Hemodynamic instability (excludes bradycardia), **Both:**
 - Finding, $\geq$ **One:**
 - Systolic blood pressure, $\geq$ **One:**
 - < 90 mmHg
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽³²⁾
 - Labile ⁽³³⁾
 - MAP < 65 mmHg
 - Intervention, $\geq$ **One:**
 - Antiarrhythmic
 - Electrical cardioversion ^(9, 20)
 - Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾
 - Hemodynamic monitoring, invasive ⁽³⁴⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽⁵⁴⁾

- $\text{FiO}_2 > 50\%$ (0.50)
- Respiratory intervention every 1-2h ⁽⁵⁶⁾
- Temporary pacemaker insertion performed within last 2d ⁽³¹⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One:** ⁽⁵⁷⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Arrhythmia controlled or resolved ^(40, 41)
 - Medication regimen established and tolerated
 - Vital signs within acceptable limits
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For atrial arrhythmia use Episode Day 3 at a higher level of care
 - For a condition other than atrial arrhythmia use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Arrhythmia controlled or resolved ⁽⁴¹⁾
 - Medication regimen established and tolerated
 - Vital signs within acceptable limits
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁴³⁾
 - **Complication or comorbidity and clinically stable for discharge, One:**
 - Heart failure resolving or at baseline ⁽³²⁾
 - **Partial Responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Antiarrhythmic medication adjustment (includes PO) ⁽⁴²⁾
 - **Unfractionated heparin, continuous and, $\geq$ One:** ⁽⁴⁴⁾
 - Antiphospholipid antibody syndrome ⁽⁴⁵⁾
 - Bleeding, active or increased risk
 - Creatinine clearance ≤ 30 mL/min ⁽⁴⁶⁾
 - BMI ≥ 40 kg/m² or weight > 120 kg ⁽⁴⁷⁾
 - **Heart failure and, Both:** ⁽⁴⁸⁾
 - **Symptom or finding, $\geq$ One:** ⁽³²⁾
 - O_2 sat $\leq 91\%$ (0.91) and $<$ baseline ⁽³²⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽⁴⁹⁾
 - **Intervention, $\geq$ One:** ⁽⁵⁰⁾
 - **Diuretic, One:** ⁽⁵¹⁾
 - ≥ 2 x/24h
 - Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline ^(32, 42)
 - Dialysis
 - Post Critical care ≤ 24 h
 - **INTERMEDIATE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Arrhythmia controlled or resolved ⁽⁴¹⁾
 - Medication regimen established and tolerated
 - O_2 sat within acceptable limits ⁽⁵⁸⁾
 - Vital signs within acceptable limits
 - **Partial Responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - **Atrial arrhythmia and, $\geq$ One:** ⁽⁵²⁾
 - Antiarrhythmic

- Conversion from IV to PO antiarrhythmic with no proarrhythmic risk $\leq 24\text{h}$
- Conversion from IV to PO antiarrhythmic with proarrhythmic risk $\leq 3\text{d}$ ^(26, 53)
- NIPPV ⁽⁵⁴⁾
- Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽⁵⁵⁾
- Permanent pacemaker placement within last 24h ⁽¹⁵⁾
- Post cardioversion and post Critical care $\leq 24\text{h}$
- Initiation or dose adjustment of antiarrhythmic (includes PO) with proarrhythmic risk $\leq 3\text{d}$ and, $\geq$ **One:** ^(24, 25, 26)
 - Dofetilide ⁽²⁷⁾
 - Sotalol

● **CRITICAL, $\geq$ One:**

● Hemodynamic instability, **Both:**

● Finding, $\geq$ **One:**

● Systolic blood pressure, $\geq$ **One:**

- < 90 mmHg
- < 110 mmHg with chronic hypertension
- > 30 mmHg decrease from baseline ⁽³²⁾
- Labile ⁽³³⁾
- MAP < 65 mmHg

● Intervention, $\geq$ **One:**

- Antiarrhythmic
- Electrical cardioversion ^(9, 20)
- Digoxin toxicity and administration of digoxin-specific antibody ⁽²⁹⁾

- Hemodynamic monitoring, invasive ⁽³⁴⁾

- Mechanical ventilation

● NIPPV and, $\geq$ **One:** ⁽⁵⁴⁾

- $\text{FiO}_2 > 50\%(0.50)$
- Respiratory intervention every 1-2h ⁽⁵⁶⁾

- Temporary pacemaker insertion performed within last 2d ⁽³¹⁾

- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 4, One:**

● **ACUTE, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾

- Arrhythmia controlled or resolved ⁽⁴¹⁾
- Medication regimen established and tolerated
- Vital signs within acceptable limits

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽⁴³⁾

● **Complication or comorbidity and clinically stable for discharge, One:**

- Heart failure resolving or at baseline ⁽³²⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

● **INTERMEDIATE, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾

- Arrhythmia controlled or resolved ⁽⁴¹⁾
- Medication regimen established and tolerated
- O_2 sat within acceptable limits ⁽⁵⁸⁾
- Vital signs within acceptable limits

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**

- **Use Extended Stay subset**

- Refer for secondary review

● **CRITICAL, One:**

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁵⁹⁾

● **HOME, Both:**

- Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁶⁰⁾
 - Patient or caregiver demonstrates ability to manage care
- Arrhythmia discharge planning, **All:**
 - Education on the importance of, **All:**
 - Strict medication adherence
 - Participating in an exercise program
 - Avoiding tobacco, excessive alcohol, and illicit drug use
 - Understanding when and where to seek help
 - Provide comprehensive written discharge and teaching instructions ⁽⁶¹⁾
 - Medication reconciliation ⁽⁶²⁾
 - Follow-up care planned ⁽⁶³⁾
 - Identify and address transportation needs

● **HOME CARE, Both:**

- Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁶⁰⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(64, 65)
- Skilled services, $\geq$ **One:** ⁽⁶⁶⁾
 - Skilled nursing ⁽⁶⁷⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁶⁸⁾
- Arrhythmia discharge planning, **All:**
 - Education on the importance of, **All:**
 - Strict medication adherence
 - Participating in an exercise program
 - Avoiding tobacco, excessive alcohol, and illicit drug use
 - Understanding when and where to seek help
 - Provide comprehensive written discharge and teaching instructions ⁽⁶¹⁾
 - Medication reconciliation ⁽⁶²⁾
 - Follow-up care planned ⁽⁶³⁾
 - Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

- Level of care appropriateness, **One:**
 - Support systems available ⁽⁶⁹⁾
- Skilled treatment, $\geq$ **One:**
 - Clinical assessment ⁽⁷⁰⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁷¹⁾

● **SKILLED MEDICAL OR THERAPY, Both:**

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq$ 1x/wk
 - Treatment precluded in a lower level
- Services, $\geq$ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁶⁷⁾
- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(72, 73)
 - Functional impairments requiring at least supervision ⁽⁷⁴⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁶²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁶⁷⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(72, 73)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁴⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁶²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(72, 73)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁴⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁶²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(72, 73)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁴⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁷⁵⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁶²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level

- In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One**:
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁷⁶⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(77, 78)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷⁹⁾
- Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁶²⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
308 CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITH MCC	n/a	3.5	CMS GMLOS
309 CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITH CC	n/a	2.3	CMS GMLOS
310 CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITHOUT CC/MCC	n/a	1.8	CMS GMLOS
ATRIAL FIBRILLATION	33%	2.6	InterQual
ATRIAL FLUTTER	28%	3	InterQual
BRADYCARDIA	44%	2.8	InterQual
SUPRAVENTRICULAR TACHYCARDIA	46%	2.5	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

In addition to an initial evaluation and determining whether rhythm or rate control is required, patients with new-onset atrial fibrillation (AF) or atrial flutter should also be considered for anticoagulation therapy based on risk factors and validated scoring systems that determine annual stroke risk. The choice of agent is based on cost, tolerability, patient risk factors, bleeding risk, drug interactions, clinical characteristics, patient preference and shared decision-making. Preventative treatment for stroke is irrespective of the type of AF (paroxysmal, persistent, long-standing persistent, or permanent) (Joglar et al., J Am Coll Cardiol 2023:).

In nonvalvular heart disease, the CHA₂DS₂-VASC score is the most validated scoring tool to estimate the annual thromboembolic/stroke risk of a patient with atrial fibrillation and to determine the need for anticoagulation. A higher score corresponds to a higher risk of stroke. Therapeutic anticoagulation is recommended when the annual stroke risk is high or $\geq 2\%$ (0.02) per year. However, when the annual risk for stroke is intermediate or $\geq 1\%$ (0.01) to $< 2\%$ (0.02), discussions regarding anticoagulation may be beneficial as part of shared decision-making between the provider and the patient (Joglar et al., J Am Coll Cardiol 2023:). Some risk factors increasing the odds of stroke or systemic embolism in patients with AF or atrial flutter include older patients or those assigned female sex at birth, compromised kidney function, history of stroke, or transient ischemic attack (TIA) (Ntaios et al., Int J Stroke 2017, 12: 589-96). The CHA₂DS₂-VASC score can be calculated by adding the points associated with each risk factor for a maximum score of 9:

Congestive heart failure = 1

Hypertension = 1

Age ≥ 75 years = 2

Diabetes mellitus = 1

Stroke/TIA/thromboembolic event = 2

Vascular disease (prior MI, angina pectoris, PAD, or coronary artery bypass surgery) = 1

Age 65-74 years = 1

Sex category, female = 1

Indications for long-term anticoagulation include (Joglar et al., J Am Coll Cardiol 2023:):

- AF or flutter and previous stroke, TIA, or CHA₂DS₂-VASC score ≥ 2 in men or ≥ 3 in women ($\geq 2\%$ (0.02)/year stroke risk): dabigatran, rivaroxaban, apixaban, edoxaban, warfarin with a targeted INR of 2.0-3.0
- AF or flutter and CHA₂DS₂-VASC score 1 in men or 2 in women ($\geq 1\%$ (0.01) to $< 2\%$ (0.02)/year stroke risk): based on shared decision-making, direct oral anticoagulants (DOACs) may be reasonable
- AF or flutter and CHA₂DS₂-VASC score 0: anti-thrombotic therapy is of no benefit

Guidelines recommend DOACs such as apixaban, rivaroxaban, edoxaban, and dabigatran over warfarin for the prevention of stroke in patients with nonvalvular AF or mechanical heart valve. They are administered in a fixed dose, daily or twice daily, without requiring laboratory anticoagulation intensity monitoring. The risk for intracranial hemorrhage was reduced with DOACs compared with vitamin K antagonists. The risk for subdural hematoma is higher with the use of vitamin K antagonists compared with antiplatelet agents. These medications should not be used in patients with prosthetic heart valves or hemodynamically significant valve disease (e.g., mitral valve stenosis) or advanced liver disease (e.g., Child-Pugh C). Continued inpatient stay is not required to monitor drug levels for DOACs. In select patients with a risk of major bleeding from long-term anticoagulation, a non-pharmacologic approach to stroke prevention may include percutaneous left atrial appendage occlusion or left atrial appendage exclusion or excision by cardiac surgery. Compared with warfarin and DOACs, left atrial appendage occlusion procedure is non-inferior and a cost-effective treatment for preventing stroke or thromboembolism in patients with atrial fibrillation. However, those patients undergoing surgical left atrial appendage excision/exclusion with a CHA₂DS₂-VASC score of ≥ 2 would still require adjunctive therapeutic anticoagulation (Joglar et al., J Am Coll Cardiol 2023;; January et al., Circulation 2019, 140: e125-e51; Lip,

International journal of clinical practice 2019, 73: e13285; Zhu et al., International journal of cardiology 2017, 227: 436-42; Touma et al., Am J Cardiol 2015, 115: 533-41; Bloom et al., Am J Cardiol 2014, 113: 1066-74; Cameron et al., BMJ Open 2014, 4: e004301; Connolly et al., Stroke 2014, 45: 1672-8; Harel et al., J Am Soc Nephrol 2014, 25: 431-42; January et al., Circulation 2014, 130: 2071-104; Lega et al., PLoS One 2014, 9: e91398; Lega et al., J Thromb Haemost 2014, 12: 337-43; Ruff et al., Lancet 2014, 383: 955-62; Sardar et al., J Am Geriatr Soc 2014, 62: 857-64).

5:

Atrial fibrillation is rapid, irregular, incomplete contractions of the atria accompanied by an irregular ventricular response. Atrial flutter is rapid, regular contractions of the atria (usually 250-400 beats per minute) accompanied by regular or irregular ventricular response (usually 140-150 beats per minute) depending on the degree of block.

6:

Patients with new-onset atrial fibrillation (AF) may require admission for initial evaluation, stabilization, and anticoagulation initiation depending on risk(s). In patients with known AF, the need for admission is based upon the patient's symptomatic tolerance of the arrhythmia and/or initiation of treatment with antiarrhythmic agents for rhythm control, which warrants close monitoring in a hospital setting based on potential electrocardiogram sequelae.

The clinical diagnosis of AF or atrial flutter is confirmed by electrocardiogram (ECG). In addition to ECG testing, as part of an initial evaluation, patients with new onset AF should also undergo transthoracic echocardiography to assess left ventricular and left atrial dimension and function. Laboratory evaluation should include thyroid, renal, and hepatic function tests as well as electrolyte and complete blood count measurement (January et al., Circulation 2019, 140: e125-e51; January et al., Circulation 2014, 130: 2071-104).

7:

The Observation level of care is appropriate for patients with paroxysmal or persistent atrial fibrillation or flutter who remain hemodynamically stable, requiring cardiac monitoring and intervention to achieve rate control. In addition, shared decision-making should be made to discuss the best strategies for managing AF based on comorbid conditions, medications, and long-term benefits (Joglar et al., J Am Coll Cardiol 2023;; Mace, Observation Medicine Principles and Protocols, First edition / ed. 2017). Stability is demonstrated by a heart rate of less than 110 beats per minute and a systolic blood pressure greater than or equal to 100 mmHg (Gaddy et al., Emerg Med Clin North Am 2017, 35: 549-69).

8:

Instruction: Treatment is inclusive of interventions provided in a medical practitioner's office, outpatient clinic (e.g., Urgent Care center), or emergency department.

9:

Electrical cardioversion for atrial fibrillation (AF) should only be performed if the confirmed onset is within 48 hours or there is no prior history of stroke or transient ischemic attack (TIA) within 6 months. Patients with a history of valvular disease are contraindicated for electrical cardioversion, and those with electrolyte abnormalities or digoxin toxicity should be managed with rate control only (Atzema and Singh, Cardiol Clin 2018, 36: 141-59). Electrical cardioversion is also indicated in hemodynamically unstable patients caused by AF or those that are hemodynamically stable but failed pharmacological conversion or performed as the initial therapy for rhythm control (Joglar et al., J Am Coll Cardiol 2023:).

10:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with

diverse expertise in varied geographic settings.

11:

Instruction: These criteria pertain to the patient who has a known history of atrial arrhythmia.

12:

Ibutilide is an antiarrhythmic agent used for the pharmacological conversion of atrial fibrillation and flutter. However, with this drug, there could be potential complications, including fatal arrhythmias such as torsades de pointes, and thus, patients should be closely monitored. Ibutilide should be avoided in patients with a prolonged QT interval, severe hypokalemia, or an ejection fraction less than 40%(0.40). These patients are also at high risk for non-sustained ventricular arrhythmia. Intravenous administration of magnesium sulfate prior to treatment with ibutilide may decrease the risk of QT prolongation, which often results in fatal arrhythmia.

Instruction: Cardiac monitoring is appropriate following ibutilide administration. Patients who are hemodynamically stable can be monitored in Observation. The Critical level of care is appropriate if the patient becomes symptomatic or hemodynamically unstable (January et al., Heart Rhythm 2019, 16: e66-e93).

13:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

14:

Digoxin is indicated in patients with heart failure (HF) and reduced left ventricular function. Although digoxin slows ventricular response, its longer onset of action (>1 hour) and peak effect time (approximately 6 hours after administration) make it a less than optimal choice when rapid rate control is desired. It is commonly used in patients with HF since it is one of the few medications used for rate control that does not have negative inotropic effects. Adverse effects of digoxin administration include ventricular arrhythmias, atrioventricular (AV) block, or sinus node dysfunction. In older patients, patients with renal dysfunction, and in patients who are taking amiodarone, propafenone, or calcium channel blockers, dose adjustment based on serum levels is warranted (Nishimura et al., Circulation 2017, 135: e1159-e95).

15:

Pacemaker implantation in the setting of atrial fibrillation (AF) is often related to underlying symptomatic bradycardia which could be a result of sick sinus syndrome. While dual chamber pacing has been shown to significantly reduce the recurrence of AF in these patients, overdrive pacing algorithms inconsistently react to premature atrial complexes. Permanent pacemaker implantation is not recommended for the treatment of AF in the absence of other indications for pacemaker implantation (January et al., Heart Rhythm 2019, 16: e66-e93).

16:

Heart failure is a comorbidity of atrial fibrillation or atrial flutter.

17:

New York Heart Association (NYHA) classification for heart failure is defined as follows (Yap et al., Clinical cardiology 2015, 38: 621-8; New York Heart Association, Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis. 1964):

- Class I - No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.
- Class II - Slight limitation of physical activity. Comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.
- Class III - Marked limitation of physical activity. Comfortable at rest. Less than ordinary physical activity causes fatigue, palpitation, dyspnea, or anginal pain.
- Class IV - Inability to carry on any physical activity without discomfort. Symptoms of cardiac insufficiency or of anginal syndrome may be present even at rest. If any physical activity is undertaken,

discomfort is increased.

18:

Supraventricular Tachycardia (SVT) is a general umbrella term for any supraventricular arrhythmia and may include atrioventricular nodal tachycardia, atrial tachycardia (e.g., multifocal atrial tachycardia), paroxysmal atrial tachycardia (PAT), and Wolfe-Parkinson-White (WPW) syndrome to name a few.

19:

Intravenous (IV) antiarrhythmic medication should be used to restore sinus rhythm in hemodynamically stable patients with supraventricular tachycardia after vagal maneuvers (e.g., Valsalva, carotid massage, or facial immersion in cold water) have failed. Adenosine and nondihydropyridine calcium channel antagonists are recommended because of their rapid onset and short half-life (Brugada et al., European heart journal 2019:).

20:

Electrical cardioversion (synchronized) is indicated for acute therapy of supraventricular tachycardias in patients who have failed conservative management such as vagal maneuvers or when medications such as adenosine have failed to resolve the arrhythmia and the patient is unstable hemodynamically. In addition, synchronized electrical cardioversion is acutely indicated when patients remain hemodynamically stable but have failed initial pharmacological therapy or have contraindications (Page et al., Circulation 2016, 133: e506-74).

21:

Information: The term bradycardia is used to define patients with a heart rate of less than 60 beats per minute. This may not be problematic or require treatment unless it becomes symptomatic. Typical symptoms of bradycardia may include weakness, dizziness, fatigue, dyspnea, syncope, or confusion.

22:

Symptoms of bradycardia include:

- Dizziness
- Lightheadedness
- Presyncope
- Syncope
- Confusion

23:

Bradycardia is a heart rate less than 60 beats per minute.

24:

Instruction: Application of this criteria point applies to patients who require hospitalization for cardiac monitoring while sotalol or dofetilide dose is being initiated or adjusted. Sotalol and dofetilide may prolong the QT interval, potentially precipitating torsades de pointes.

25:

Initial refers to the first time a medication or treatment is utilized. During the period of initiation, if the medication or treatment is temporarily discontinued up to 24 hours (therapeutic pause) it is still considered initial. If a tolerated medication or treatment is discontinued for more than 24 hours and then restarted, it is not considered initial. Medications that were initiated on an outpatient basis and are discontinued prior to hospitalization may be considered initial based on the need for monitoring or duration of discontinuation. Medications that were temporarily held due to concurrent illness and are resumed are not considered initial.

26:

Drugs with a high proarrhythmic risk may cause worsening of an existing arrhythmia or precipitate a new one. Cardiac monitoring is recommended during initiation of these medications (January et al., Heart Rhythm 2019, 16: e66-e93).

27:

Dofetilide is a class III antiarrhythmic used for the pharmacological conversion of atrial fibrillation/flutter to normal sinus rhythm. It selectively blocks potassium channels, which results in the widening of the QRS complex, thereby potentially increasing the risk for QT prolongation and development of torsades de pointes. These patients should be monitored closely for any arrhythmias. The risk can be minimized by adjusting the dose based on creatinine clearance since it is renally excreted. Dofetilide should not be initiated outside of the hospital, and recent guidelines recommend a 3-day minimum hospital stay for those patients being initiated, reinitiated, or dose adjusting on dofetilide (Joglar et al., J Am Coll Cardiol 2023;; January et al., Heart Rhythm 2019, 16: e66-e93).

28:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

29:

For hemodynamically unstable patients, or those with lethal arrhythmia, immunotherapy with digoxin-specific antibodies (known as Fab fragments) may be administered via an intravenous (IV) bolus to bind intravascular free digoxin. A complete response usually occurs within 4 hours. In patients with heart failure, digoxin-specific antibodies should be used with caution. Patients should be monitored for signs of volume overload and hypokalemia due to movement of intravascular potassium into the cell (Kusumoto et al., Circulation 2019, 140: e382-e482).

30:

Isoproterenol should be used with caution in patients with suspected coronary ischemia because it causes an increase in myocardial oxygenation demand (Kusumoto et al., Circulation 2019, 140: e382-e482).

31:

Temporary pacemaker placement is recommended for patients with symptomatic bradycardia or heart block (Kusumoto et al., Circulation 2019, 140: e382-e482; Oken et al., JAMA cardiology 2019:).

32:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

33:

Blood pressure is considered labile when it is extremely variable and difficult to control.

34:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

35:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

36:

Instruction: The criteria point "Comorbid atrial fibrillation or flutter in a hospitalized patient" can only be applied to patients who have had atrial fibrillation or atrial flutter as a complication of their current inpatient stay in an acute care facility. The criteria point cannot be applied for patients who have been either discharged to home or transferred to a post-acute facility.

37:

Acute atrial fibrillation (AF) is defined when atrial fibrillation is noted during an acute care hospitalization or secondary to an acute illness. It also includes AF detected for the first time during a patient's hospitalization for another condition. Acute AF is associated with longer lengths of stay and higher morbidity and mortality, including predisposing a patient to an increased risk of recurrent AF. Acute AF as a complication has been implicated in broad range of medical conditions and could be associated with certain surgeries (cardiac and non-cardiac). Incidence of acute AF ranges between 1%(0.01) and almost 50%(0.5) in the medical setting, especially for patients admitted with severe sepsis or septic shock or those having non-cardiac surgery (e.g., lung resection, thoracic surgery). When acute AF occurs, patients should be evaluated for possible triggers or substrates that may warrant further workup as an inpatient. Examples of triggers include (but are not limited to) inflammation, electrolyte imbalance, infection, bleeding, increased procedural times, and volume overload (Chyou et al., Circulation 2023, 147: e676-e98).

38:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights. For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

39:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

40:

Patients discharged from Observation should receive outpatient evaluation with echocardiogram (ECG) and early follow-up with their cardiologist within 3 days of discharge (Atzema and Singh, Cardiol Clin 2018, 36: 141-59).

41:

Stability is demonstrated by a resting heart rate less than 110 beats per minute and a systolic blood pressure greater than or equal to 100 mmHg (Hindricks et al., European Heart Journal 2020, 42: 373-498; Gaddy et al., Emerg Med Clin North Am 2017, 35: 549-69).

42:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

43:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

44:

While low molecular weight heparin (LMWH) generally is considered first-line treatment for patients with symptomatic thromboembolic disease when anticoagulation is indicated, there are specific clinical circumstances in which unfractionated heparin is preferable, including (Stevens et al., Chest 2021, 160: e545-e608; Ortel et al., Blood

Adv 2020, 4: 4693-738):

- Renal failure (CrCl < 30 mL/min)
- Morbid obesity (BMI ≥ 40 kg/m²) or weight > 120 kg
- Antiphospholipid antibody syndrome
- Bleeding

Risk of bleeding, including, but not limited to:

- Anticipated surgical intervention or interventional procedure (e.g., hemodynamic instability, thrombolysis)
- Anticipated need for urgent discontinuation

45:

Thrombotic events are common in patients with antiphospholipid antibody syndrome. The ease of use and lack of need for monitoring have led to interest in using direct oral anticoagulants (DOACs) for management, but warfarin continues to be preferred by many providers. While the use of low molecular weight heparin (LMWH) in bridging to warfarin may be appropriate for some patients, unfractionated heparin is preferred in patients with known or suspected catastrophic antiphospholipid syndrome, or in patients with anticoagulant-refractory disease who may require higher intensity dosing (and as such, would incur an increased risk of bleeding) (Cohen et al., J Thromb Haemost 2021, 19: 892-908).

46:

Patients with creatinine clearance less than or equal to 30 mL/min should be treated with unfractionated heparin when parenteral anticoagulation is necessary. Certain direct oral anticoagulants and low molecular weight heparin should be dose adjusted in presence of renal failure. Bridging to warfarin with medication and lab monitoring is necessary for at least 5 days until the INR is therapeutic for two consecutive days (Stevens et al., Chest 2021, 160: e545-e608; Konstantinides et al., Eur Heart J 2020, 41: 543-603)

47:

Unfractionated heparin is often used in the management of acute thromboembolic events in patients with a body mass index (BMI) ≥ 40 kg/m² or weight > 120 kg; there is growing evidence of the use of direct oral anticoagulants (DOACs) in this specific patient population based on a large study. Previously, the 2016 International Society on Thrombosis and Haemostasis (ISTH) guidelines recommended against the use of DOACs in patients with a BMI ≥ 40 kg/m² or weight > 120 kg, and a 2019 systematic review (19 studies, including pharmacokinetic, studies, phase III RCTs and post hoc analyses, and retrospective cohort studies) continued to suggest caution (Kido et al., Pharmacotherapy 2020, 40: 72-83). However, a recent meta-analysis (5 observational studies, n=6585) concluded that these agents were non-inferior to warfarin (Elshafei et al., J Thromb Thrombolysis 2021, 51: 388-96). Furthermore, a large study of over 36000 patients with atrial fibrillation and stage III obesity (e.g., BMI ≥ 40 kg/m²) showed a relative risk reduction in stroke by 25%(0.25) with the use of DOACs compared to warfarin, 57%(0.57) relative risk reduction in significant bleeding and 34%(0.34) reduction in all-cause mortality. However, patients who have undergone bariatric surgery were not well represented in studies comparing warfarin and DOACs, and for these patients, warfarin may be a better option as drug levels can be monitored (Joglar et al., J Am Coll Cardiol 2023;).

48:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

49:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

50:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias,

continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., *Circulation* 2017, 136: e273-e344).

51:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., *Circulation* 2022, 145: e895-e1032; Maddox et al., *J Am Coll Cardiol* 2021, 77: 772-810; Rosendorff et al., *Circulation* 2015).

52:

Hemodynamic stability is evidenced by adequate tissue and organ perfusion. Blood pressure and heart rate are parameters commonly used to assess hemodynamic stability. Clinical findings associated with hemodynamic stability include normal peripheral pulses, warm extremities, adequate urine output, and neurological stability. The Critical level of care is appropriate for patients who develop hemodynamic instability.

53:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

54:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., *PulmCCM* 2018; Nardi et al., *F1000Res* 2017, 6: 290; Osadnik et al., *Cochrane Database Syst Rev* 2017, 7: CD004104; Allison and Winters, *Emerg Med Clin North Am* 2016, 34: 51-62; Gregoretti et al., *Crit Care Clin* 2015, 31: 435-57).

55:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula****Room air** 21%(0.21)**1L** 24%(0.24) **4L** 36%(0.36)**2L** 28%(0.28) **5L** 40%(0.40)**3L** 32%(0.32) **6L** 44%(0.44)**Simple oxygen masks**

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

56:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

57:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

58:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

59:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

60:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g.,

secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

61:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

62:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

63:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

64:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home

would be cumbersome and places the patient at risk

- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

65:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

66:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

67:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

68:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the

need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

69:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

70:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

71:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

72:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

73:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

74:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

75:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

76:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

77:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

78:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive

- Inspiratory muscle weakness

79:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.